3 Ancient Greek Mathematics

3.1 Overview of Ancient Greek Civilization & Early Philosophy

Ancient Greek civilization dates from around 800 BC, centered on the peninsula between the Adriatic Sea (to the west) and the Aegean (to the east) that forms the mainland of the modern Greek state. There was no central controlling empire. Semi-independent city-states were connected by trade and culture, but largely governed themselves. Accomplished seafarers, the Greeks extended their reach round the northern and eastern coasts of the Mediterranean, from Iberia (Spain) to the Black Sea, Anatolia (western Turkey), and Egypt, building and capturing a network of city-states and trading outposts.

Greek Territory around 500 BC

Philip of Macedon 'unified' the Greek peninsula just before his death in 336 BC. His son, Alexander the Great, subsequently launched a massive military campaign conquering Persia, Egypt, Babylon, and western India; he died in Babylon in 323 BC. Alexander left various governors to manage his conquered territories: some of these political structures endured for centuries (Egypt's Ptolemaic dynasty), whereas others were overthrown after only a few years (India/Pakistan). While Alexander's conquests did not produce a long-lasting centralized Greek empire, they were effective at expanding the reach of Greek culture and philosophy, and brought external ideas into the Greek tradition.

The core part of Greek territory was absorbed by the Roman empire around 146 BC. In line with typical Roman practice, those who agreed to accept Roman governors and taxation could eventually become Roman citizens.[7] As such, the Greek culture of inquiry and scholarship staid largely intact under Roman rule. Greek learning was central to the later Byzantine (eastern Roman) empire centered on Byzantium/Constantinople/Istanbul, which lasted until the capture

[7] While this might sound reasonable, resistance wasn't realistic, and a significant period of enslavement might first have to be endured...

of Constantinople by the Islamic Ottomans in 1453. For centuries prior, Islamic scholarship had itself been significantly influenced by the ancient Greeks. On consequence of the fall of Constantinople was the exodus of scholars to Rome, helping to fuel the nascent European Renaissance.

Early Greek Thought

Greek mathematics is part of a much wider philosophy encompassing a change of emphasis from practicality to abstraction. One reason for this was the Greek blending of religion/mysticism with natural philosophy: a desire to describe the natural world while preserving the (seeming) perfection and logic in the gods' design.

Early Greek inquiry into natural phenomena involved the personification of nature (e.g., sky = man, earth = woman). By 600 BC, philosophers were attempting to describe natural phenomena in terms of logical structures. For example, some viewed matter as being comprised of the 'four elements' (fire, earth, water, air) combined in the correct proportions. While the system of the world was considered divinely-designed, explanations relying on the whims of the gods came to be discouraged.

While the Greeks certainly used mathematics for practical purposes, philosophers idealized logic and were unhappy with approximations. This led to the development of *axiomatics, theorems* and *proof*, concepts for which there is scant pre-Greek evidence. The ancient Greek language is indeed the source of three words of critical importance:

Mathematics *Mathematos* (μαθήματος) meant knowledge or learning. The term covered essentially anything that might be taught in Greek schools. The stem *math* (knowledge) is preserved in other modern usage: e.g. *polymath.*

Geometry Literally *earth-measure*, this is a combination of two terms:

- *Gi* (γη) Dates from pre-5th century BC, meaning *land, earth* or *soil.* Capitalized (Γη) it could refer to the *Earth* (as a goddess).
- *Metron* (μέτρον) A *weight* or *measure*, a *dimension* (length, width, etc.), or the *metre* (rhythm) in music.

Theorem From *theoreo* (θεώρέω), meaning 'I contemplate/consider.' In a mathematical context this become *theorema* (θεώρημα): a proposition to be proved.

Greek Education and Scholarship

Ancient Greece had several schools, mostly private and open only to men. Typically arithmetic was taught until age 14, followed by geometry and astronomy until age 18.

The most famous scholars of ancient Greece were the Athenian trio of Socrates, Plato and Aristotle, each of whom taught his successor,[8] and whose writings became central to both Western and Islamic philosophical tradition. Plato's *Academy* in Athens provided the model for centuries of schooling; the centrality of geometry to Plato's curriculum was evidenced by the famous inscription above its entryway: "Let none ignorant of geometry enter here."

[8] The birth of Socrates to the death of Aristotle covered 470–322 BC.

Ancient Greek Enumeration

The Greeks had two primary forms of enumeration.

Attic Greek The Attic system (Attica refers to the area around Athens) dates from 7-800 BC. Strokes were used for 1–4, with larger numerals using the first letter of the words for 5, 10, 100, 1000 and 10000. For example,

- Πεντε (*pente*) means five, so Π denoted the number 5.
- Δεκα (*deca*) means ten, so Δ represented 10.
- H (*hekaton*), X (*khilias*) and M (*murias*) denoted 100, 1000 and 10000 respectively.
- 𐅄, 𐅅 and 𐅆 represented 50, 500, and 5000 respectively.
- Combinations represented larger numbers: e.g., X𐅅HHΠI= 1706. This is similar to both Egyptian hieroglyphs and Roman numerals. Unlike the latter, Attic Greek did not use pre-subtraction: thus 9 was written ΠIIII instead of IΔ.

In modern Greece, Attic numerals are still used occasionally for formal purposes, similarly to how other western countries might use Roman numerals (e.g. construction date of a building).

Ionic Greek Ionia refers to the Anatolian coast. This system dates from around 400 BC and uses the Greek alphabet, an approach possibly copied from Egyptian hieratic enumeration. Ionic numerals slowly displaced the older Attic system.

The Ionic alphabet included several unfamiliar symbols whose shapes and usage changed over time: *digamma* (Ϝ), *stigma* (ϛ), *qoppa* (ϙ, ϟ), and *sampi* (ͳ, ϡ).

Larger numbers were depicted in various ways. A left subscript mark like a comma denoted thousands and/or M (with superscripts) for multiples of 10000. For example,

$$35268 = {}_{,}\lambda{}_{,}\varepsilon\sigma\xi\eta = \overset{\gamma}{\mathrm{M}}{}_{,}\varepsilon\sigma\xi\eta$$

1	α	10	ι	100	ρ
2	β	20	κ	200	σ
3	γ	30	λ	300	τ
4	δ	40	μ	400	υ
5	ε	50	ν	500	φ
6	Ϝ, ϛ	60	ξ	600	χ
7	ζ	70	ο	700	ψ
8	η	80	π	800	ω
9	θ	90	ϙ, ϟ	900	ͳ, ϡ

The Greeks also employed Egyptian fractions, denoting reciprocals with a *keraia*: e.g., ¼ = δ′ along with a few special symbols like ∠′ for ½ (perhaps related to the special Egyptian hieroglyph 𓐛 for the same). The use of Egyptian fractions dominated in Europe well into the middle ages.

Both the Attic and Ionic systems were suitable for record-keeping but terrible for calculations! Later Greek mathematicians adapted the Babylonian sexagesimal system with Ionic notation, helping cement its modern use of in astronomy, navigation and time-keeping.
Ionic enumeration is still used in modern Greece, though Hindu–Arabic numerals are now more typical. By the middle-ages, a bar was placed over numerals to distinguish them from words, while modern practice is to insert a *keraia* similarly to how the ancients denoted fractions: e.g. $89 = \overline{\pi\theta} = \pi\theta'$.

Exercises 3.1. As with other ancient numerical systems, what matters is their structure (positional, decimal, etc.), not the details. Don't memorize the ancient Greek alphabet!

1. State the number 1896 in both Attic and Ionic notation.
2. Represent $\frac{8}{9}$ as a sum of distinct unit fractions (Egyptian style). Express the result in (Ionic) Greek notation.

   (*The answer to this problem is non-unique*)
3. For tax purposes, the ancient Greeks would approximate the area of a quadrilateral field by multiplying the averages of the two pairs of opposite sides. In one example, the two pairs of opposite sides were given as

   $$\begin{aligned} a &= \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \frac{1}{32} \quad \text{opposite} \quad c = \frac{1}{8} + \frac{1}{16}, \quad \text{and,} \\ b &= \frac{1}{2} + \frac{1}{4} + \frac{1}{8} \quad \text{opposite} \quad d = 1 \end{aligned}$$

   where the lengths are in fractions of of a *schonion*, a measure of approximately 150 feet. Find the average of a and c, the average of b and d, and thus the approximate area of the field in square *schonion.* The taxman then rounds up the answer to collect a little more tax!

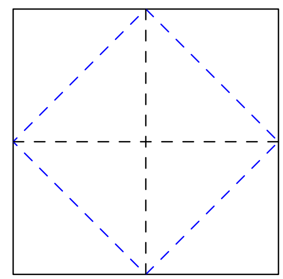
3.2 Pre-Euclidean Greek Mathematics

By subsuming much that came before it, the publication of Euclid's *Elements* (c. 300 BC) forms a natural breakpoint in Greek mathematics. In this section, we consider the contributions of several pre-Euclidean mathematicians. Very few sources exist from before 400 BC, so almost everything is inferred from the writings and commentaries of others.[9]

Thales of Miletus (c. 624–546 BC)

Thales is credited as one of the first people to state general abstract principles. A trader based in Miletus, a city-state in Anatolia, Thales travelled widely and was likely exposed to mathematical and philosophical ideas from all round the Mediterranean. Here are several statements at least partly attributed to him:

- The angles at the base of an isosceles triangle are equal.
- Any circle is bisected by its diameter.
- A triangle inscribed in a semi-circle is right-angled (still known as Thales' Theorem).



Thales' major development is *generality*: his propositions concern *all* triangles, *all* circles, etc. Babylonian & Egyptian mathematics had *examples*, but there is scant indication of abstract generality. In line with the Greek idea of a theorem ('to look at'), Thales' reasoning was likely pictorial.

As an example of contemporary geometric reasoning, by 425 BC Socrates could describe how to halve (or double) the area of a square by joining the midpoints of edges and folding.

Pythagoras of Samos (c. 572–497 BC)

Samos is an island in the Aegean, not far from Miletus. Like Thales, Pythagoras also travelled widely, eventually settling in Croton, southern Italy, where he founded a school—this endured a century after his death. Plato is believed to have learned much of his mathematics from a Pythagorean named Archytas.

The Pythagoreans practiced a mini-religion whose ideas lay outside the mainstream of Greek society.[10] One of their mottos, "All is number," emphasised their belief in the centrality of pattern and proportion. The following quote[11] gives some flavor of the Pythagorean way of life.

> After a testing period and after rigorous selection, the initiates of this order were allowed to hear the voice of the Master [Pythagoras] behind a curtain; but only after some years, when their souls had been further purified by music and by living

[9] For instance, most of our knowledge of Socrates comes from the voluminous writings of Plato and Aristotle. The earliest known Greek textbook/compilation was written around 430 BC (*Elements of Geometry* by Hippocrates of Chios); no copy survives, though most of its material probably made it into Book I of Euclid.

[10] They believed in the transmigration of souls, were vegetarians, and accepted women as students; controversial ideas indeed!

[11] Van der Waerden, *Science Awakening* pp 92–93.

> in purity in accordance with the regulations, were they allowed to see him. This purification and the initiation into the mysteries of harmony and of numbers would enable the soul to approach [become] the Divine and thus escape the circular chain of re-births.

Musical harmony The relationship between music and number was of great interest to the Pythagoreans. They are credited with relating musical intervals to ratios of lengths of vibrating strings:

- Identical strings whose lengths are in the ratio 2:1 vibrate an *octave* apart.
- A *perfect fifth* corresponds to the ratio 3:2.
- A *perfect fourth* corresponds to the ratio 4:3.

The use of these ratios to tune musical instruments is still known as *Pythagorean tuning.*[12]

Mathematical results attributable to the Pythagoreans Theorems 21–34 in Book IX of Euclid's *Elements* are Pythagorean in origin. For instance:

Theorem. **(IX.21)** *A sum of even numbers is even.*

(IX.27) *Odd less odd is even.*

The Pythagoreans also studied perfect numbers: those which equal the sum of their proper divisors (e.g., $6 = 1 + 2 + 3$). They seem to have observed the following famous result.

Theorem (IX.36). *If $2^n - 1$ is prime then $2^{n-1}(2^n - 1)$ is perfect.*

They also considered square and triangular numbers ($\frac{1}{2}m(m+1)$) and tried to express geometric shapes as numbers; all of this was in line with their belief that all matter could be formed from basic shapes.

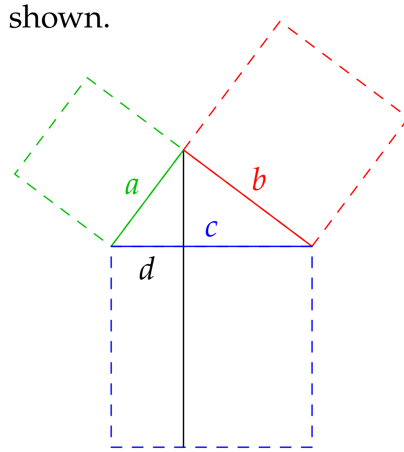
Of course, the most famous result associated with Pythagoras is the theorem that bears his name.

Theorem (I.47). *The square on the hypotenuse of a right-triangle equals (has the same area as) the sum of the squares on the remaining sides.*

This result is the capstone of Book I or Euclid's *Elements*; indeed the book seems to have been structured precisely to provide a rigorous justification. It is generally believed that the Pythagoreans did not possess a rigorous proof, though here is an inferred argument that would have been in line with their approach.

[12] This contrasts with the more modern *equal temperamnent* tuning, where the frequency ratios of all semitone intervals are identical (twelfth-root of 2). Equal temperament is mathematically challenging because it requires the accurate computation of this ratio.

Inferred proof. Label the side-lengths of the right-triangle a, b, c, as shown.
Drop the altitude to the hypotenuse c.
Let d be the length of the part of the hypotenuse underneath a.
The large triangle comprises two smaller similar triangles, whecne



$$\begin{aligned} a : d = c : a &\implies a^2 : ad = cd : ad \\ &\implies a^2 = cd \end{aligned}$$

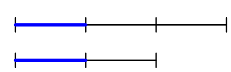
The square on a therefore has the same area as the rectangle below the left side of the hypotenuse.
Repeat the calculation on the other side to obtain $b^2 = c(c - d)$, and sum to complete the proof. ■

From a modern viewpoint, there is nothing wrong with this argument. So why did Euclid feel the need to devote an entire book to re-proving it?! The problem lies with the Pythagoreans' contradictory use of the ancient Greek concepts of number and length...

Ratios and Incommensurability: length versus number

As with other ancient cultures, *number* to the Greeks meant *positive integer.* While the Greeks certainly used fractions, these were properly considered ratios of integers. Indeed ratios were used to *compare* physical quantities such as length and volume, as well as non-physical quantities like number. For instance, we'd happily say that the integers 8 and 4 are in the ratio $2:1$.

Definition. Let m, n be positive integers. Lengths (or other physical quantities) are said to be in the ratio $m:n$ if some sub-length divides exactly m times into the first and n times into the second.
Lengths are *commensurable* if some sub-length divides exactly into both, and *incommensurable* otherwise.



For instance, the pictured rods have commensurable lengths in the ratio $\ell_1 : \ell_2 = 3 : 2$, since the largest possible common sub-length goes three times into the first rod and twice into the second.

While modern mathematics has no problem with *irrational ratios* (e.g., the diagonal of a square to its side is $\sqrt{2}:1$), this conflicts with the Pythagorean belief that *any two lengths were commensurable.* Identifying lengths with real numbers (underlined), we restate this assertion in modern language:

$$\forall \underline{m}, \underline{n} \in \mathbb{R}^+,\ \exists \underline{\ell} \in \mathbb{R}^+,\ a, b \in \mathbb{N},\ \text{such that } \underline{m} = a\underline{\ell} \text{ and } \underline{n} = b\underline{\ell}$$

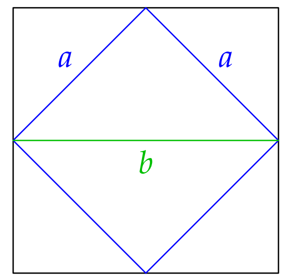
But this is utter nonsense, for it claims that every ratio of real numbers $\underline{m} : \underline{n} = a : b$ is *rational*!
The Pythagorean commensurability supposition stems from their basic tenets: all is number (including length ratios), and the design of the gods is perfect (number means positive integer).

The discovery of incommensurable ratios produced something of a crisis for the Pythagoreans; a possibly apocryphal story states that a disciple named Hippasus (c. 500 BC) was set adrift at sea as punishment for its revelation.

By 340 BC, the Greeks were happy to state that incommensurable lengths exist.

Theorem (Aristotle). *If the side and diagonal of a square are commensurable, then odds equal evens.*

Inferred proof. In Socrates' doubled-square, suppose that side and diagonal of the smaller square are in the ratio $a : b$; note that a and b are integers!



Assume (without loss of generality) at least one of the integers is odd, for otherwise the common sub-length may be doubled. The larger square is twice the area of the smaller, whence the square numbers have ratio

$$b^2 : a^2 = 2 : 1$$

Plainly b^2 is even and so also is b. Writing $b = 2k$, we have

$$4k^2 : a^2 = 2 : 1 \implies a^2 : k^2 = 2 : 1$$

whence a^2, and thus a, is also even. Whichever of a, b was odd is also even: contradiction! ∎

Note the similarity of this argument to the modern proof of the irrationality of $\sqrt{2}$.

This discussion helps to illuminate the importance of geometry to the Greeks. Lengths could be anything (in essence they are continuous), whereas numbers are more limited: numbers cannot describe the relationship between all lengths.

Returning to the Pythagoreans' inferred proof of their own theorem, in the ancient Greek context, the argument only makes sense if the numbers a, b, c, d are *integer multiples of a common sub-length*. Aristotle's result shows that such a sub-length need not exist.

Theaetetus of Athens (417–369 BC) Likely the source of much of the most difficult parts (Books VII & X) of the *Elements*, Theaetetus' approach to (in)commensurability comes from applying what is now known as the Euclidean algorithm to lengths (line segments).

Definition. Repeatedly apply the division algorithm to lengths $\underline{a} > \underline{b}$ ($\underline{a}$ is longer than $\underline{b}$):

$$\begin{aligned} \underline{a} &= q_1\underline{b} + \underline{r}_1 & \underline{r}_1 &< \underline{b} & &(\exists q_1 \in \mathbb{N}_0 \text{ and a length } \underline{r}_1 < \underline{b}) \\ \underline{b} &= q_2\underline{r}_1 + \underline{r}_2 & \underline{r}_2 &< \underline{r}_1 \\ \underline{r}_1 &= q_3\underline{r}_2 + \underline{r}_3 & \underline{r}_3 &< \underline{r}_2, \quad \text{etc.} \end{aligned}$$

We say that $\underline{a}$ and $\underline{b}$ are *commensurable* if the algorithm terminates: some remainder $\underline{r}_n$ divides exactly into $\underline{r}_{n-1}$. Otherwise $\underline{a}$ and $\underline{b}$ are *incommensurable*.

Ratios are *equal* $\underline{a} : \underline{b} = \underline{c} : \underline{d}$ precisely when the sequences of quotients in the algorithm are equal.

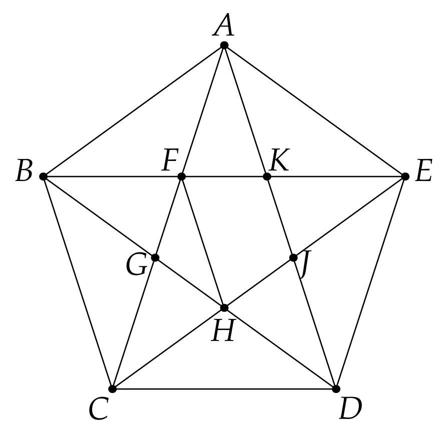
If $\underline{a}$ and $\underline{b}$ are commensurable, then $\underline{r_n}$ is their *greatest common sub-length*. If we write $\underline{a} = a\underline{r_n}$ and $\underline{b} = b\underline{r_n}$ for some integers a, b and rewrite the algorithm in modern fashion, we obtain the standard Euclidean algorithm demonstration of $\gcd(a, b) = 1$.

Example 1 The algorithm applies to pairs of integers (start with commensurable rods!).

$$\begin{aligned} \underline{37} &= 2 \cdot \underline{13} + \underline{11} \\ \underline{13} &= 1 \cdot \underline{11} + \underline{2} \\ \underline{11} &= 5 \cdot \underline{2} + \underline{1} \\ \underline{2} &= 2 \cdot \underline{1} \end{aligned} \qquad\qquad \begin{aligned} \underline{148} &= 2 \cdot \underline{52} + \underline{44} \\ \underline{52} &= 1 \cdot \underline{44} + \underline{8} \\ \underline{44} &= 5 \cdot \underline{8} + \underline{4} \\ \underline{8} &= 2 \cdot \underline{4} \end{aligned}$$

Since we obtain the same sequence of quotients $(2, 1, 5, 2)$, we conclude that $37 : 13 = 148 : 52$.

Example 2 We outline an argument showing that the diagonal $\underline{a} = \overline{AC}$ and side $\underline{b} = \overline{AB}$ of a regular pentagon $ABCDE$ are incommensurable.



1. $\triangle BAG$ is isosceles (count angles!), whence $|AB| = |AG|$.
2. The first line of the algorithm reads
$$|AC| = |AG| + |GC| = |AB| + |GC|$$
so we write $\underline{a} = q_1\underline{b} + \underline{r_1}$ where $r_1 = |GC|$ and $q_1 = 1$.
3. Since $|GC| = |AF|$, the second line of the algorithm reads
$$|AB|\,\big(= |AG|\,\big) = |AF| + |FG| = |GC| + |FG|$$
so that we again have quotient $q_2 = 1$.
4. Prove that $\triangle DCG \cong \triangle EHF$. But then $|GC| = |FH|$ is the diagonal of the interior regular pentagon. The third line of the algorithm ($|GC|$ divided by $|FG|$) is therefore the same as the first: dividing the diagonal to the side of a regular pentagon. The algorithm therefore continues forever with all quotients being 1.

Example 3 In modern language, the diagonal to the side of a square is the incommensurable ratio $\sqrt{2} : 1$. We apply Theaetetus' algorithm:

$$\begin{aligned} \underline{\sqrt{2}} &= 1 \cdot \underline{1} + (\underline{\sqrt{2} - 1}) \\ \underline{1} &= 2 \cdot (\underline{\sqrt{2} - 1}) + (\underline{3 - 2\sqrt{2}}) \end{aligned}$$

Observe that $3 - 2\sqrt{2} = (\sqrt{2} - 1)^2$, whence the second line reads $1 = 2x + x^2$. The following lines in the algorithm may therefore be obtained by repeatedly multiplying by x:

$$x = 2x^2 + x^3, \quad x^2 = 2x^3 + x^4, \quad \text{etc.},$$

which produces a never-ending sequence[13] of quotients: $1, 2, 2, 2, 2, 2, \ldots$

[13] If you like number theory, investigate the relationship of Theaetetus' algorithm to continued fractions...

Eudoxus of Knidos (c. 390–337 BC) One of the most prolific pre-Euclidean mathematicians, Eudoxus attended and perhaps taught at Plato's academy. He is famous for explaining how to calculate with ratios of lengths (segments).

Definition. $A : B > C : D$ if there exist positive integers m, n such that $mA > nB$ and $mC \leq nD$.

At first glance it appears as if Eudoxus is telling us how to compare *rational* numbers. Indeed, if A, B, C, D are integers, then the definition may be rewritten:

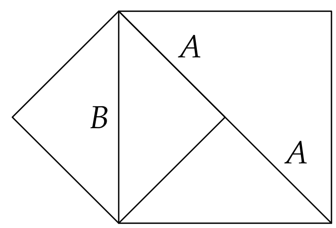
$$\frac{A}{B} > \frac{n}{m} \geq \frac{C}{D}$$

is trivially satisfied by taking $m = D$ and $n = C$. To Eudoxus, however, A, B, C, D could be interpreted as *segments*. Building on the work of Theaetetus, his mathematics told the Greeks how to approximate incommensurable ratios with rational ones.

Example 1 To see that $13 : 3 > 17 : 4$, simply choose $m = 4$ and $n = 17$ to obtain

$$4 \cdot 13 = 52 > 51 = 3 \cdot 17$$

Example 2 We show that the ratio of the side to the diagonal of a square is greater than $1 : 2$; equivalently, the diagonal is less than twice the side.[14]



As pictured, side : diagonal $= A : B$. Choosing $m = D = 2$ and $n = C = 1$, we see that

$$mA = 2A > B = nB \qquad \text{(diag > side of large square)}$$

In modern language, this is merely $\frac{1}{\sqrt{2}} > \frac{1}{2}$.

Arguments about Infinity: Zeno of Elea (c. 450 BC)

The Greeks provided some of the first discussions (and disagreements) about infinity and infinitesimals. Zeno's arguments, in particular, have provided philosophical fodder for millennia, illustrating some of the difficulties behind modern calculus. Here are two of the most famous:

Achilles and Tortoise Achilles chases a Tortoise. After time t_0, Achilles reaches the Tortoise's starting position but the Tortoise has moved on. After a second time interval t_1, Achilles reaches the Tortoise's second position; again the Tortoise has moved. In this manner Achilles spends a total time $t_0 + t_1 + t_2 + \cdots$ chasing the Tortoise. Zeno's paradoxical conclusion is that Achilles never catches the Tortoise.

This paradox may be resolved (see Exercise 8), at least if we assume both runners travel at constant speeds. Though it be split into infinitely many subintervals of time, the total duration of the chase can expressed as the (finite) value of a convergent *infinite series.*

[14] This is trivial by the triangle inequality, though it is interesting to see it in Eudoxus' context.

Arrow paradox An arrow is shot from a bow. At any instant the arrow doesn't move. If time is made up of instants, then the arrow never moves.

This time Zeno debates the the idea that a positive time interval can be considered as a sum of infinitesimal instants. This is the essential philosophical difficulty of *integration.*

Zeno's arguments were marshaled against Galileo (early 1600s) and both Newton and Leibniz (late 1600s/early 1700s) as their methods of infinitesimal calculus became widely used.

Constructions and Geometry

By the mid 5th century BC, Greek mathematicians were solving geometric problems using *ruler-and-compass* (peg-and-cord) constructions. This approach might have come to Greece from India, or possibly arose organically. Constructions were based on three rules, which became the first three postulates (axioms) of Book I of Euclid's *Elements.*

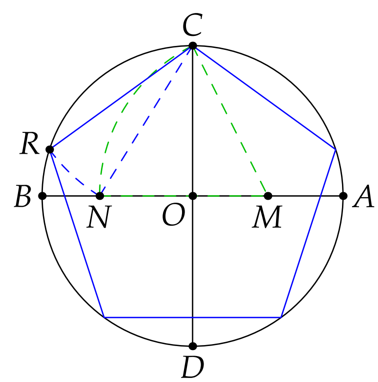
1. Two points may be joined by a straight line (segment).
2. A segment may be extended indefinitely.
3. Given a center and radius, one may draw a circle.

Theorems were often stated as *problems*: e.g., *to bisect a given angle.* A proof provided first a *construction,* then an argument justifying that the construction really had solved the problem.

By the time of Euclid, the Greeks knew many constructions, including those of an equilateral triangle, a square and a regular pentagon in a given circle.

Example We construct a regular pentagon in a given circle.[15]

1. Draw perpendicular diameters $\overline{AB}$, $\overline{CD}$ and bisect $\overline{OA}$ at M.
2. Draw an arc centered at M with radius $|CM|$. Let N be the intersection of this arc with $\overline{OB}$.
3. Draw an arc centered at C with radius $|CN|$. Let R be the intersection of this arc with the original circle.
4. Move $|CR|$ around the circle to create a regular pentagon.



A purely geometric proof of the validity of this construction is too difficult for us, but you can easily check it using a calculator, Pythagoras' and the cosine rule: if the circle has radius 2, then

$$\begin{aligned}|CR|^2 = |CN|^2 &= |ON|^2 + |OC|^2 \\ &= (\sqrt{5}-1)^2 + 2^2 = 10 - 2\sqrt{5} \\ &= 2^2 + 2^2 - 2 \cdot 2 \cdot 2\cos 72^\circ\end{aligned}$$

[15]Theorem IV. 11 of the *Elements* presents a less practical construction. Ours follows from Theorem XIII. 10: if a regular pentagon, hexagon and decagon are inscribed in a circle, then their sides form a right-triangle.

The appearance of $\sqrt{5}$ in this discussion relates to the golden ratio $\frac{1+\sqrt{5}}{2}$ which is in fact the ratio of the diagonal to the side of the pentagon: exactly what Theaetetus was calculating on page 27. Pentagons and pentagrams were viewed as mystical by many ancient Greeks.

Construction problems have motivated mathematicians ever since. In 1796, Carl Friedrich Gauss (then only nineteen) constructed a regular 17-gon. A full classification of the constructable regular polygons came in 1837.

Theorem. *A regular n-gon is constructable (using only ruler-and-compass constructions) if and only if $n = 2^k F_1 \cdots F_r$ where $F_1, \ldots, F_r$ are distinct primes of the form $2^{(2^n)} + 1$.*

After the 17-gon, the next prime-sided constructable n-gon has $257 = 2^{2^3} + 1$ sides!

By 400 BC, the Greeks were referencing the second and third *impossible constructions of antiquity*:

1. Trisecting a general angle.
2. Doubling (the volume of) a given cube.
3. Squaring a circle (construct a square with the same area as a given circle).[16]

It wasn't until the advent of field theory in the 1800s that such were proved to be impossible using ruler-and-compass constructions.

Summary

Several of the mathematical techniques in this section are difficult and the results technical. It isn't important to become proficient with all of these ideas. Play with them to help you develop an appreciation of two overarching points:

1. Even before Euclid, the focus of Greek mathematics was more abstract and less practical than that of other ancient cultures (Egypt, Babylon, China, etc.), largely due to the influence of wider Greek philosophy and religion. The modern liberal arts ideal of learning for its own sake—to celebrate the beauty of knowledge and expand the mind—is, to a large extent, a Greek inheritance.
2. The ancient Greeks pondered fundamental mathematical questions and concepts—*number* versus *length*, continuity, irrationality, infinitesimals, constructions—ideas that have stimulated mathematical research ever since. These particular issues would not be resolved rigorously until the 1800s, when luminaries such as Gauss, Cauchy and Riemann developed modern analysis and algebra.

[16]"You can't square that circle," is a metaphor for something impossible, or beyond human comprehension.

Exercises 3.2. *Key concepts: Abstraction, Ratios of magnitudes (number ≠ length), Incommensurability (irrationality), Difficulties reasoning with infinity, Ruler-and-compass constructions*

1. Construct five Pythagorean triples using the formula $\left(n, \frac{n^2-1}{2}, \frac{n^2+1}{2}\right)$ where n is odd. Construct five more using the formula $\left(m, \left(\frac{m}{2}\right)^2 - 1, \left(\frac{m}{2}\right)^2 + 1\right)$ where m is even.

2. Suppose $2^n - 1 = p$ is prime (its only positive divisors are itself and 1). List the positive divisors of $2^{n-1}(2^n - 1)$ and hence prove Theorem IX. 36.

3. Draw a *picture* with dots to show that eight times a triangular number plus 1 makes a square, and that any odd square minus 1 is eight times a triangular number. That is:

   (a) $8 \cdot \frac{1}{2}m(m+1) + 1$ is a perfect square.

   (b) If n is odd, then $n^2 - 1 = 8 \cdot \frac{1}{2}m(m+1)$ for some m.

4. Find a ruler-and-compass construction to bisect a given angle and show that it is correct.

5. Sketch a construction inscribing a regular hexagon in a circle.

   (*Assume you can construct an equilateral triangle on a given segment—Thm I. 1 of Euclid*)

6. (A line-doubling paradox) One line has twice the length of another and so has *more* points. However, there is a bijective correspondence between the points on these lines; the two lines therefore have the *same number* of points.

   Explain the second observation. How can you resolve the paradox?

7. The *cycle of fifths* is a musical concept stating that twelve perfect fifths equals seven octaves (pg. 24). State this claim *numerically*, and show that it is a contradiction.

   (*Hint: two strings are seven octaves apart if their lengths are in the ratio* $2^7 : 1$)

8. Recall Zeno's paradox of Achilles and the Tortoise. Suppose Achilles travels at speed v_A, the Tortoise at speed $v_B < v_A$, and that the tortoise starts a distance d ahead of Achilles.

   (a) Prove that $t_n = \frac{d}{v_A}\left(\frac{v_T}{v_A}\right)^n$ for each positive integer n.

   (b) Compute $\sum_{n=0}^{\infty} t_n$ using the geometric series formula from calculus.

   (c) Verify the time-value computed in (b) by considering the motion of Achilles relative to the Tortoise.

9. Use Theaetetus' definition of equal ratios to prove that $46 : 6 = 23 : 3$.

10. (Hard) A line of length 1 is divided at x so that $\frac{1}{x} = \frac{x}{1-x}$. Prove that 1 and x are incommensurable. Moreover, show that $1 : x$ is *the same* as diagonal : side of a regular pentagon.

    (*Hint: the first line of the algorithm is* $1 = x + x^2 \ldots$)

11. (Hard) Let $a > b$ and c be *positive lengths.* Use Eudoxus' definition to *prove* that $c : b > c : a$.

    (*Hint: let n be the smallest integer such that $n(a-b) \geq c$; its existence is the "archimedean property"*)